Challenging Mock · Paper 1

Question 1

An integrable function f satisfies

$$f(x+1) = 2 - f(x)$$

for every real x, and

$$\int_0^1 f(x)\,dx = 3.$$

Find

$$\int_0^4 x f(x)\,dx.$$

A 8

B 0

C 2

D -8

E 4

Challenging Mock · Paper 1

Question 2

When $x^{100}+1$ is divided by

$$x^2+5x+4,$$

what is the remainder?

A $\dfrac{1-4^{100}}{3}x+\dfrac{7-4^{100}}{3}$

B $\dfrac{1-4^{100}}{3}x+\dfrac{4^{100}+5}{3}$

C $\dfrac{4^{100}-1}{3}x+\dfrac{7-4^{100}}{3}$

D 2

E $\dfrac{4^{100}-1}{3}x+\dfrac{4^{100}+5}{3}$

Challenging Mock · Paper 1

Question 3

For $k > -4$, the curves

$$y = x^3 - 4x \qquad \text{and} \qquad y = kx$$

enclose two bounded regions. When k is increased by 2, the total enclosed area doubles. Find k.

A $2\sqrt{2} - 4$

B $2\sqrt{2} - 2$

C $\sqrt{2} - 2$

D $2 - \sqrt{2}$

E $2\sqrt{2}$

Challenging Mock · Paper 1

Question 4

Positive real numbers a and b satisfy $a + b = 12$. What is the maximum possible value of

$$ab(a - b)^2$$

A 1296

B 1728

C 2304

D 864

E 1152

Challenging Mock · Paper 1

Question 5

A robot travels from $A = (1, 2)$ to $B = (8, 4)$ in two straight segments. It must cross a scanner at $P = (t, 0)$ on the positive horizontal axis, and the scanner works only for $t \geq 5$. What is the shortest possible total distance?

A $\sqrt{85}$

B $3\sqrt{10}$

C $4 + \sqrt{26}$

D $2 + \sqrt{53}$

E $5 + 2\sqrt{5}$

Challenging Mock · Paper 1

Question 6

A sequence has period 4. Its first four terms are $1, 2, 3, 4$ in an unknown order, and $a_{n+4} = a_n$. Given that

$$\sum_{n=1}^{20} n a_n = 540,$$

find $a_1 a_4 - a_2 a_3$.

A -8

B 2

C 8

D 0

E -2

Challenging Mock · Paper 1

Question 7

Evaluate the infinite series

$$\frac{1}{2}+\frac{2}{2^2}+\frac{1}{2^3}+\frac{4}{2^4}+\frac{1}{2^5}+\frac{6}{2^6}+\cdots,$$

where every odd-position numerator is 1, while the numerator in position $2m$ is $2m$.

A 2

B $\frac{22}{9}$

C $\frac{14}{9}$

D $\frac{10}{9}$

E $\frac{16}{9}$

Challenging Mock · Paper 1

Question 8

Three circles with radii 2, 3 and 10 are pairwise externally tangent. Their three points of pairwise tangency are A, B and C.

What is the area of triangle ABC?

A $\dfrac{60}{13}$

B 6

C 30

D $\dfrac{30}{13}$

E 5

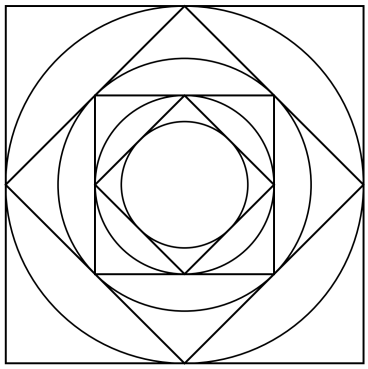
Challenging Mock · Paper 1

Question 9

A circle is inscribed in a square of side 4. A new square is then inscribed in that circle. The process is repeated indefinitely.

At every stage, record the area inside the current square but outside its inscribed circle. What is the sum of all the recorded areas?

Paper 1, Question 9



Boundary diagram only; the regions whose areas are recorded are not shaded.

A $6(4-\pi)$

B $4(4-\pi)$

C $16-2\pi$

D $32-4\pi$

E $8(4-\pi)$

Challenging Mock · Paper 1

Question 10

The real numbers p, q, r satisfy

$$p + q + r = 0 \qquad \text{and} \qquad p > r > 0.$$

How many solutions does

$$p\sin^2\theta + q\sin\theta + r = 0$$

have for $0 < \theta < \pi$?

A 4

B 2

C 3

D 1

E 0

Challenging Mock · Paper 1

Question 11

A circle has centre $(p, 6)$, is tangent to the x-axis, and passes through $(4, 10)$. There are two possible values of p. What is the distance between them?

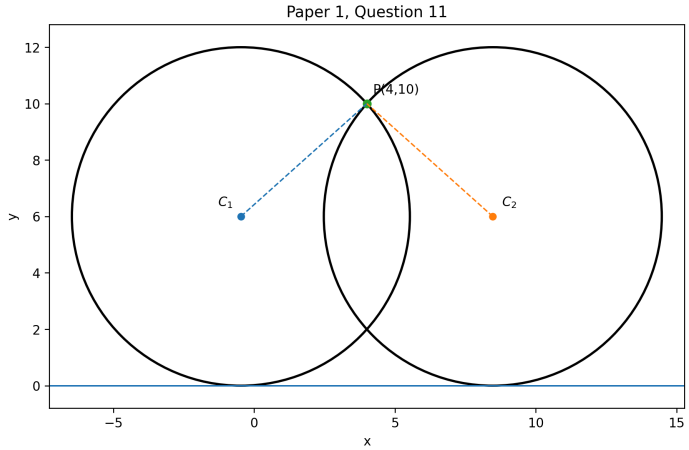


Diagram for Paper 1, Question 11 — not to scale.

A $2\sqrt{5}$

B $4\sqrt{5}$

C 10

D $6\sqrt{5}$

E 8

Challenging Mock · Paper 1

Question 12

A function f has exactly three real zeros: $-3, 1, 5$. Define

$$g(x) = |f(2|x| - 1)|.$$

How many distinct real zeros does g have?

A 5

B 2

C 3

D 6

E 4

Challenging Mock · Paper 1

Question 13

How many six-digit positive integers can be formed using six distinct digits selected from

$$0, 1, 2, 3, 4, 5, 6$$

and are divisible by 15?

A 480

B 432

C 528

D 552

E 456

Challenging Mock · Paper 1

Question 14

A line $y = mx + c$ passes through $(2, 5)$ and meets the positive coordinate axes, forming a triangle of area 25 with the axes. There are two possible gradients.

What is the product of these two gradients?

A 5

B $\frac{5}{4}$

C $\frac{25}{4}$

D 25

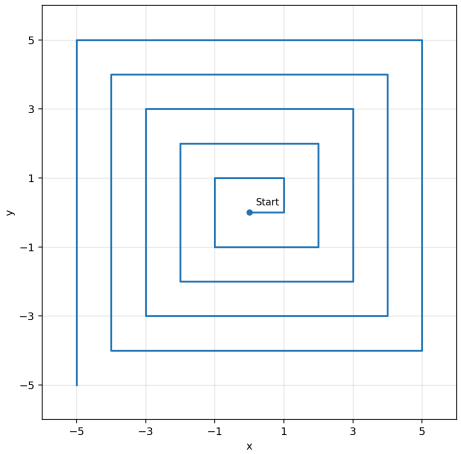
E $\frac{25}{2}$

Challenging Mock · Paper 1

Question 15

A path begins at $(0, 0)$. It moves right 1, up 1, left 2, down 2, right 3, up 3, and continues in this pattern.

Thus each positive integer length occurs on two consecutive segments, and the directions repeat right, up, left, down. What is the straight-line distance from the origin after 20 segments?



Schematic path; the final endpoint is deliberately not labelled.

A $5\sqrt{2}$

B $10\sqrt{2}$

C 5

D 10

E 20

Challenging Mock · Paper 1

Question 16

A quadratic function f satisfies $f(0) = 2$ and $f(4) = 18$. The ordinates at $x = 0, 1, 2, 3, 4$ are used with the trapezium rule, with strip width 1, to estimate

$$\int_0^4 f(x)\,dx.$$

The estimate exceeds the exact integral by 2. Find $f(2)$.

A -6

B 6

C 10

D -2

E 2

Challenging Mock · Paper 1

Question 17

For which values of the real parameter k does

$$x^4 - 2(k+1)x^2 + k^2 = 0$$

have exactly four distinct real solutions?

A $k > 0$

B $k > -\frac{1}{2}$

C $-\frac{1}{2} < k < 0$

D $k \neq 0$

E $(-\frac{1}{2}, 0) \cup (0, \infty)$

Challenging Mock · Paper 1

Question 18

Find the maximum value of

$$x + \sqrt{(x+1)(9-x)}$$

for $-1 \le x \le 9$.

A $4 + 3\sqrt{2}$

B $9 + \sqrt{2}$

C 9

D $4 + 5\sqrt{2}$

E $5\sqrt{2}$

Challenging Mock · Paper 1

Question 19

Let $k \geq 0$. For which values of k does

$$\big|\,|x^2 - 4x| - 3\,\big| = k$$

have exactly six distinct real solutions?

A $0 < k < 1$

B $1 < k < 3$

C $1 \leq k < 3$

D $k > 3$

E $k = 2$

Challenging Mock · Paper 1

Question 20

The parabola $y = x^2$ and the line $y = mx + 4$ enclose an area of 36. Find m^2.

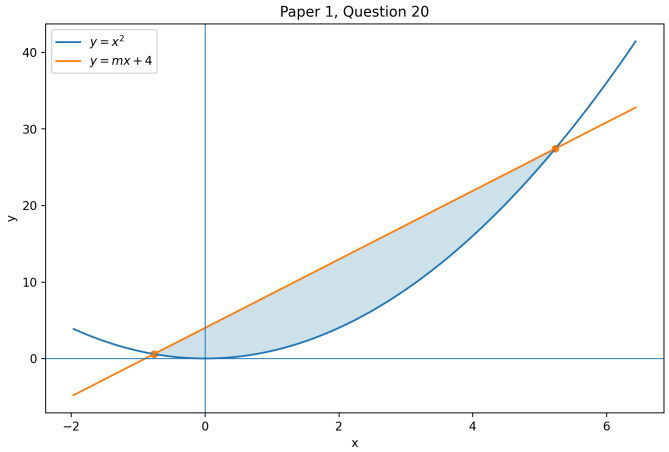


Diagram for Paper 1, Question 20 — not to scale.

A 24

B 32

C 20

D 16

E 18

Challenging Mock · Paper 2

Question 1

A real function f has exactly four distinct real zeros

$$a < b < c < d.$$

The equation

$$f(x^2 - 4) = 0$$

has exactly five distinct real solutions. Which statement must be true?

A $a < b < -4 < c < d$

B $a = -4 < b < c < d$

C $a < -4, \quad b = -4, \quad -4 < c < d$

D $-4 < a < b < c < d$

E $a < -4 < b < c < d$

Challenging Mock · Paper 2

Question 2

A student claims that $x^2 < y^2$ implies $x < y$, using the following argument:

Line 1. $x^2 - y^2 < 0$.
Line 2. $(x - y)(x + y) < 0$.
Line 3. Exactly one of $x - y$ and $x + y$ is negative.
Line 4. Since $x - y < x + y$, the negative factor is $x - y$.
Line 5. Hence $x - y < 0$, so $x < y$.

Which is the first incorrect line?

A Line 2

B Line 4

C Line 1

D Line 3

E Line 5

Challenging Mock · Paper 2

Question 3

Let f be a real polynomial. Every real root of f is simple, and $f(0)f(1) \neq 0$. Let k be the number of real roots of f in $(0,1)$. Consider

$$P:\ f(0)f(1) < 0, \qquad Q:\ k \text{ is odd.}$$

Which describes the relationship between P and Q?

A P is necessary but not sufficient for Q

B P is necessary and sufficient for Q

C P is neither necessary nor sufficient for Q

D P is sufficient but not necessary for Q

E Q is impossible

Challenging Mock · Paper 2

Question 4

Let a, b, c be positive real numbers. Consider

$$P:\ a, b, c \text{ are the sides of a nondegenerate triangle,}$$

$$Q:\ (a+b+c)^2 > 2(a^2+b^2+c^2).$$

Which is correct?

A P is necessary and sufficient for Q

B Q is true for all positive a, b, c

C P is sufficient but not necessary for Q

D P is necessary but not sufficient for Q

E P is neither necessary nor sufficient for Q

Challenging Mock · Paper 2

Question 5

A differentiable function f satisfies

$$f'(x) < 0 \qquad \text{for every } x \in [0,1].$$

Consider:

I. $\displaystyle\int_0^{1/2} f(x)\,dx > \int_{1/2}^{1} f(x)\,dx,$

II. $\displaystyle\int_0^{1} f(x)\,dx < \int_0^{1} \big(f(x)\big)^2\,dx,$

III. $f(0) < f(1).$

Which statements must be true?

A I only

B I and III only

C II only

D III only

E I and II only

Challenging Mock · Paper 2

Question 6

A real sequence (a_n) satisfies the statement

For every real M, there exists a positive integer N such that, for every $n \geq N$, $a_n > M$.

Which is the logical negation of this statement?

A There exists a real M such that, for every positive integer N and every $n \geq N$, $a_n \leq M$.

B For every real M, there exists a positive integer N such that, for every $n \geq N$, $a_n \leq M$.

C For every real M and every positive integer N, there exists $n \geq N$ with $a_n \leq M$.

D There exists a real M and a positive integer N such that, for every $n \geq N$, $a_n \leq M$.

E There exists a real M such that, for every positive integer N, there exists $n \geq N$ with $a_n \leq M$.

Challenging Mock · Paper 2

Question 7

Let

$$N = 2^a 3^b 5^c,$$

where a, b, c are nonnegative integers. Which condition is necessary and sufficient for 5400 to divide N^2?

A $180 \mid N$

B $360 \mid N$

C $90 \mid N$

D $120 \mid N$

E $60 \mid N$

Challenging Mock · Paper 2

Question 8

Every integer is coloured exactly one of red or green. For every integer n:

- if n is red, then $n + 2$ is green;
- if n is green, then $n + 100$ is red.

Given that 205 is green, which integer must be green?

A 9

B 109

C 54

D 7

E 105

Challenging Mock · Paper 2

Question 9

A function f satisfies

$$0 \le f(x) \le 1 \qquad (0 \le x \le 4),$$

together with

$$\int_0^2 f(x)\,dx = 1, \qquad \int_1^3 f(x)\,dx = \frac{3}{2}.$$

What is the range of possible values of

$$\int_0^4 f(x)\,dx?$$

A $\left(\frac{3}{2}, 3\right)$

B $[2, 3]$

C $[1, 3]$

D $\left[\frac{3}{2}, 3\right]$

E $\left[\frac{3}{2}, \frac{7}{2}\right]$

Challenging Mock · Paper 2

Question 10

Let a, b, c be positive real numbers. A student claims that

$$a^2 + b^2 > c^2 \quad \Longrightarrow \quad a + b > \sqrt{2}\,c,$$

using the following argument:

Line 1. $a^2 + b^2 > c^2$.
Line 2. $2ab \le a^2 + b^2$.
Line 3. $(a+b)^2 = a^2 + b^2 + 2ab \le 2(a^2 + b^2)$.
Line 4. Therefore $(a+b)^2 > 2c^2$.
Line 5. Since $a + b$ and c are positive, $a + b > \sqrt{2}\,c$.

Which line is the first that is not justified by the preceding argument?

A Line 3

B Line 1

C Line 5

D Line 2

E Line 4

Challenging Mock · Paper 2

Question 11

Consider the quadratic

$$x^2 + ux + v.$$

Which set of conditions is necessary and sufficient for it to have two distinct roots, both lying in $(0, 1)$?

A $u < 0,\ v > 0,\ u^2 > 4v$

B $-2 < u < 0,\ v < 0,\ u^2 > 4v,\ 1 + u + v > 0$

C $-2 < u < 0,\ v > 0,\ u^2 > 4v,\ 1 + u + v > 0$

D $-2 < u < 0,\ v > 0,\ u^2 > 4v,\ 1 + u + v < 0$

E $-2 < u < 0,\ v > 0,\ u^2 \geq 4v,\ 1 + u + v > 0$

Challenging Mock · Paper 2

Question 12

A function $f : \mathbb{R} \to \mathbb{R}$ satisfies

$$f(x + y) = f(x) + f(y)$$

for all real x, y, and $f(x) \geq 0$ for every real x. Which statement must be true?

A $f(x) = x$ for every real x

B $f(x) = |x|$ for every real x

C f is strictly increasing

D $f(x) = 0$ for every real x

E No such function exists

Challenging Mock · Paper 2

Question 13

Let x be a nonzero real number. Consider

$$P:\ x \text{ is irrational}, \qquad Q:\ x+\frac{1}{x} \text{ is irrational}.$$

Which is correct?

A P is necessary but not sufficient for Q

B P is neither necessary nor sufficient for Q

C P is necessary and sufficient for Q

D Q is false for every x

E P is sufficient but not necessary for Q

Challenging Mock · Paper 2

Question 14

Let n be a positive integer. The objective is to determine whether $12 \mid n$. The following information is available:

$$\text{I.}\quad 18 \mid n^2, \qquad \text{II.}\quad 16 \mid n^3.$$

Which option is correct?

A Even the two statements together are insufficient.

B Statements I and II together are sufficient, but neither alone is sufficient.

C Either statement alone is sufficient.

D Statement II alone is sufficient.

E Statement I alone is sufficient.

Challenging Mock · Paper 2

Question 15

Let $f : \mathbb{R} \to \mathbb{R}$. Consider

$$P: \ f \text{ is one-to-one (injective)},$$

$$Q: \ f \circ f \text{ is one-to-one (injective)}.$$

Which is correct?

A P is necessary but not sufficient for Q

B P is sufficient but not necessary for Q

C P is necessary and sufficient for Q

D P is neither necessary nor sufficient for Q

E Q is impossible

Challenging Mock · Paper 2

Question 16

An integrable function f satisfies

$$f(-x) = 1 - f(x)$$

for every real x. Consider:

- I. $f(0) = \frac{1}{2}$.
- II. $\displaystyle\int_{-a}^{a} f(x)\,dx = a$ for every $a > 0$.
- III. $f(x+1) = f(x)$ for every real x.

Which statements must be true?

A I only

B I and II only

C I, II and III

D II only

E III only

Challenging Mock · Paper 2

Question 17

The side lengths of a triangle are

$$5, \qquad 5r, \qquad 5r^2.$$

What is the range of possible values of r?

A $0 < r < \dfrac{1+\sqrt{5}}{2}$

B $1 < r < \dfrac{1+\sqrt{5}}{2}$

C $\dfrac{\sqrt{5}-1}{2} \le r \le \dfrac{\sqrt{5}+1}{2}$

D $\dfrac{1-\sqrt{5}}{2} < r < \dfrac{1+\sqrt{5}}{2}$

E $\dfrac{\sqrt{5}-1}{2} < r < \dfrac{\sqrt{5}+1}{2}$

Challenging Mock · Paper 2

Question 18

Which expression is divisible by 30 for every integer n?

A $n^6 - n$

B $n^3 - n$

C $n^4 - n$

D $n^5 - n$

E $n^5 + n$

Challenging Mock · Paper 2

Question 19

Let

$$f(x) = x^2(1-x)^2.$$

Find

$$f\left(\frac{1}{2019}\right) - f\left(\frac{2}{2019}\right) + f\left(\frac{3}{2019}\right) - \cdots + f\left(\frac{2017}{2019}\right) - f\left(\frac{2018}{2019}\right).$$

A 0

B $\dfrac{2018^2}{2019^4}$

C $-\dfrac{2018^2}{2019^4}$

D $\dfrac{1}{16}$

E $\dfrac{1}{2019^4}$

Challenging Mock · Paper 2

Question 20

A polygon is called **loud** if, for each occurrence of an interior angle A, there is a *distinct angle occurrence* B in the same polygon such that

$$B = \frac{A}{2} \qquad \text{or} \qquad B = 2A.$$

The occurrence chosen as B may have the same numerical measure as A, and one occurrence may serve as the required partner for more than one occurrence of A. A polygon that is not loud is called **quiet**.

Consider polygons with interior angles

$$\begin{aligned} &\text{I.} && 30^\circ, 60^\circ, 90^\circ, \\ &\text{II.} && 40^\circ, 80^\circ, 80^\circ, 160^\circ, \\ &\text{III.} && 60^\circ, 120^\circ, 120^\circ, 120^\circ, 120^\circ. \end{aligned}$$

Which polygons are quiet?

A II only

B I only

C II and III only

D I and II only

E III only